

Neural Network Skeleton: Autonomous Technical Report

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1 Project Executive Summary

This document is an autonomous technical decomposition of the Neural Network engine. The analysis agent has parsed the source code to map the mathematical architecture and memory management strategies.

2 Engineering Foundations

The project is built on C++14 standards, focusing on: - **Manual Memory Management**: Implementation of Move Semantics for performance. - **Mathematical Precision**: Custom Linear Algebra engine for neural computations. - **Static Analysis**: Documentation generated via CI/CD pipelines.

3 Module Analysis: FirstTest.cpp

3.0.1 Identified Interface Methods

- `int main()`
- `Matrix m1(3, 3)`
- `Matrix eq1(2, 2)`
- `Matrix eq2(2, 2)`
- `assert(eq1 == eq2)`
- `assert(!(eq1 == eq2))`
- `Matrix A(2, 3)`
- `Matrix B(3, 2)`
- `Matrix H1(2,2)`
- `Matrix H2(2,2)`
- `Matrix D(2, 2)`
- `1           assert(deriv(0,1) == 0.0)`
- `0           assert(deriv(1,0) == 0.0)`
- `Matrix R(10, 10)`
- `Matrix E(5, 5)`
- `assert(false)`

4 Module Analysis: Layer.cpp

5 Module Analysis: Layer.h

5.1 Architecture: Class Layer

The agent detected the structural definition of `Layer` in this module. This component manages specific neural operations.

5.1.1 Identified Interface Methods

- `Matrix backward(const Matrix& outputGradient, double learningRate)`

6 Module Analysis: Main_Matrix_Run.cpp

6.0.1 Identified Interface Methods

- `int main()`
- `results       srand(static_cast<unsigned>(time(0)))`
- `Network net(layerSizes, 3, 0.1)`

7 Module Analysis: Matrix.cpp

7.0.1 Identified Interface Methods

- `Matrix sum_matrix(rows, cols)`
- `Matrix res(rows, cols)`

- `Matrix res(this->rows, other.cols)`
- `Matrix T(cols, rows)`
- `Matrix result(rows, cols)`
- `Matrix result(rows, cols)`
- `Matrix result(rows, cols)`
- `mt19937 gen(rd())`
- `Matrix result(rows, cols)`

8 Module Analysis: `Matrix.h`

8.1 Architecture: Class `Matrix`

The agent detected the structural definition of `Matrix` in this module. This component manages specific neural operations.

8.1.1 Identified Interface Methods

- `double& operator()(int row, int col)`
- `void randomize(double min, double max)`
- `void fillZero()`
- `void applyReLU()`

9 Module Analysis: `Network.cpp`

9.0.1 Identified Interface Methods

- `new Layer(layerSizes[i], layerSizes[i+1])`

10 Module Analysis: `Network.h`

10.1 Architecture: Class `Network`

The agent detected the structural definition of `Network` in this module. This component manages specific neural operations.

10.1.1 Identified Interface Methods

- `Matrix feedForward(const Matrix& input)`
- `void backpropagate(const Matrix& input, const Matrix& target)`
- `void train(Matrix* inputs, Matrix* targets, int numSamples, int epochs)`

11 Module Analysis: `secondtest.cpp`

11.0.1 Identified Interface Methods

- `int main()`
- `Layer myLayer(4, 10)`
- `Matrix input(4, 1)`